



Principles of Finance

Chapter13

End-of-Chapter Questions

Statistical Analysis in Finance

Multiple Choice Questions

1. A data set of salaries contains an outlier salary. The best measure of center to use for this data set is the _____.
 - A. mean
 - B. median
 - C. mode
 - D. standard deviation

Solution: B

Section 13.1 LO 2 Moderate

2. A portfolio includes shares of United Airlines stock that were purchased at different times and different prices. Which measure is best to determine the average cost of the shares of the stock?
 - A. mean
 - B. median
 - C. weighted mean
 - D. standard deviation

Solution: C

Section 13.1 LO 3 Moderate

3. Standard deviation is a measure of the _____.
 - A. center of a data set
 - B. position of a data value in a data set
 - C. area under a normal curve
 - D. spread of a distribution

Solution: D

Section 13.2 LO 1 Easy

4. How are standard deviation and variance related?
 - A. The two measures are equal to one another.
 - B. Variance is the square root of the standard deviation.
 - C. Standard deviation is the square root of the variance.
 - D. The two squared measures are equal to one another.

Solution: C

Section 13.2 LO 1 Easy

5. Which of the following is the best definition of a z-score?
- A. the distance of a data value from the mean
 - B. the number of standard deviations that a data value is from the mean
 - C. the distance of a data value from the mean divided by the sample size
 - D. the number of quartiles that a data value is from the mean

Solution: B

Section 13.3 LO 1 Easy

6. The results of a standardized test indicate that you are in the 85th percentile. What is the best interpretation of this result?
- A. You scored in the top 85% of all students taking the test.
 - B. You scored in the top 15% of all students taking the test.
 - C. Your score on the test is 85 when measured on a scale from 0 to 100.
 - D. You scored in the bottom 15% of all students taking the test.

Solution: B

Section 13.3 LO 2 Moderate

7. The interquartile range is _____.
- A. the middle 50% of a data set.
 - B. the upper 50% of a data set.
 - C. the lower 50% of a data set.
 - D. equal to the median.

Solution: A

Section 13.3 LO 2 Moderate

8. In a frequency distribution table, the sum of the relative frequencies must be equal to _____.
- A. the sample size
 - B. 1, or 100%
 - C. zero
 - D. the standard deviation of the distribution

Solution: B

Section 13.3 LO 1 Moderate

9. A change in the standard deviation of a normal distribution will result in _____.
- A. a change in the location of the peak of the curve
 - B. a change in the area under the curve
 - C. a change in the shape of the curve
 - D. a change that shifts the graph to the left or the right

Solution: A

Section 13.3 LO 2 Difficult

10. When calculating an expected value, _____.

- A. The result should always be 1.
- B. The result should always be a positive value.
- C. The result should always be a negative value.
- D. The result can be a positive or negative value

Solution: D

Section 13.5 LO 2 Moderate

11. The area under a normal curve between a z-score of -2 and a z-score of +2 is _____.

- A. 0.68
- B. 0.95
- C. 0.997
- D. dependent on the mean and standard deviation

Solution: B

Section 13.5 LO 3 Moderate

12. A scatter plot is a visualization for _____.

- A. univariate data only
- B. bivariate data only
- C. either univariate or bivariate data
- D. test scores

Solution: B

Section 13.6 LO 3 Easy

13. Which of the following is NOT a benefit of using the R statistical analysis tool?

- A. Additional features are constantly being added by the user community.
- B. It can be used on many computer platforms, including Mac, Windows, and Linux.
- C. It is free to download.
- D. Users pay an annual subscription fee.

Solution: D

Section 13.7 LO 1 Moderate

Review Questions

1. Explain the considerations that determine whether the mean or the median is the best measure of central tendency for a data set.

Solution: If outliers are present in the data set, then the mean will be distorted by the outliers and may produce a misleading result. The median is the better measure of central tendency when outliers are present in the data.

Section 13.1 LO 1 Moderate

2. Explain the difference between a mean and a weighted mean.

Solution: When calculating the mean, all data values are weighted equally. In a weighted mean, certain data values may be weighted more heavily than other data values. An example of this might be the grading criteria for a college course where the score on the final exam is weighted more heavily than a score on a quiz.

Section 13.1 LO 3 Moderate

3. Explain why the standard deviation of a data set cannot be a negative value.

Solution: When calculating the standard deviation, the deviation for each data value is calculated by subtracting the mean from the data value. Then this result is squared, and the squared deviations are added together. Then the square root of the sum of the squared deviations is calculated. When squaring a quantity, the squared result is always positive. The sum of positive results will be positive, and then the square root of a positive quantity is itself positive.

Section 13.2 LO 1 Difficult

4. Explain what a negative z-score, a positive z-score, and a z-score of zero imply.

Solution: A z-score measures the position of a value in a data set. It is calculated by subtracting the mean from the data value of interest and then dividing the difference by the standard deviation. If a z-score is negative, that indicates that the data value is less than the mean. If a z-score is positive, then the data value is greater than the mean. If a z-score is zero, then the data value is equal to the mean.

Section 13.2 LO 1 Moderate

5. Explain how quartiles can be used to detect outliers in a data set.

Solution: First, the quartiles of a data set are calculated. Then the interquartile range (IQR) is also calculated, as $IQR = Q_3 - Q_1$. Then the lower and upper bounds for outliers are calculated using the following formulas:

$$\text{Lower Bound for Outliers} = Q_1 - (1.5 \cdot IQR)$$

$$\text{Upper Bound for Outliers} = Q_3 + (1.5 \cdot IQR)$$

Any data values below the lower bound or above the upper bound are flagged as outliers, and these data values should be further investigated to determine the nature of the outlier condition.

Section 13.3 LO 3 Moderate

Problems

1. You purchased 1,000 shares of a stock for \$12 per share. Then, two months later, you purchased an additional 500 shares of the same stock at \$9 per share. Calculate the weighted mean of the purchase price for the total of 1,500 shares.

Solution:

$$\text{Weighted Mean} = \frac{1,000(\$12) + 500(\$9)}{1500} = \$11$$

Section 13.1 LO 3 Moderate

2. You score a 60 on a biology test. The mean test grade is 70, and the standard deviation is 5. Calculate and interpret your corresponding z-score.

Solution: The formula for a z-score calculation is as follows, where x is the data value, μ is the mean, and σ is the standard deviation:

$$z = \frac{x - \mu}{\sigma}$$

In this example, the data value is 60, the mean is 70, and the standard deviation is 5. The z-score is then calculated as

$$z = \frac{60 - 70}{5} = -2$$

A z-score of -2 indicates that your test grade is two standard deviations below the mean, which is a poor grade when compared to grades of the rest of the students taking the test.

Section 13.3 LO 1 Moderate

3. You score a 60 on a biology test. The mean test grade is 70, and the standard deviation is 5. What percentile does your grade correspond to?

Solution: According to the empirical rule, a z-score of -2 corresponds to the bottom 2.5% of the distribution, meaning that 97.5% of all other students had higher test grades than you.

Percentile is a measure of the area to the left of a certain data value. Thus, your test grade of 60 corresponds to the 2.5th percentile.

Section 13.3 LO 2 Difficult

4. A fast food restaurant has measured service time for customers waiting in line, and the service time follows an exponential distribution with a mean waiting time of 1.9 minutes. The restaurant has a guarantee that if customers wait in line for more than 5 minutes, their meal is free. What is the probability that a customer will receive a free meal?

Solution: To calculate the probability, find the area under the curve to the left of 5 minutes. This will be the probability that a customer will wait in line for less than 5 minutes. The mean waiting time is given as 1.9 minutes. The probability can then be calculated as follows:

$$\text{Probability} = 1 - e^{-\frac{1}{1.9}(5)} = 0.928$$

This probability of 0.928 represents the probability that a customer will wait in line for less than 5 minutes. To calculate the probability that a customer will wait in line for more than 5 minutes, subtract 0.928 from 1, which gives a probability of 0.072. Thus, there is a 7.2% chance that a customer will receive a free meal due to a wait time of longer than 5 minutes.

Section 13.4 LO 3 Difficult

5. The total value of your portfolio consists of approximately 65% stock assets, 25% bonds, and 10% cash equivalents. Historical returns have shown that stocks provide a return of 12%, bonds provide a return of 3.5%, and cash savings provide a return of 1.5%. What is the expected value of the return on this portfolio?

Solution:

$$\text{Expected Value} = 0.65(0.12) + 0.25(0.035) + 0.10(0.015) = 0.0883, \text{ or } 8.83\%$$

Section 13.5 LO2 Moderate

6. The distribution of the average annual return of the S&P 500 over a 50-year time period follows a normal distribution with a mean rate of return of 10.5% and a standard deviation of 14.3%. What is the probability that an average annual return will fall between -3.8% and 24.8%?

Solution: Note that the given interval is from one standard deviation below the mean to one standard deviation above the mean. Per the empirical rule, the probability that a data value in a normal distribution will fall within one standard deviation of the mean is 0.95.

Section 13.5 LO 3 Difficult

7. Write a short R program to find the expected return for the following data set:

Historical Return on United Airlines Stock	Associated Probability
12%	15%
5%	35%
2%	25%
-5%	14%
-10%	11%

Solution: To calculate the expected value, first create a vector in R for the historical returns, and create another vector for the associated probabilities. Then, use the weighted mean command to calculate the expected value.

The corresponding R commands would be as follows:

[R-SOFTWARE]

```
> returns <- c(12, 5, 2, -5, 10)
> probabilities <- c(0.15, 0.35, 0.25, 0.14, 0.11)
> weighted.mean(returns, probabilities)
```

[/R-SOFTWARE]

Section 13.7 LO 2 Moderate

Chapter 13 Test Bank

Statistical Analysis in Finance

These questions and problems can all be solved in a classroom setting using a handheld financial calculator.

True or False Questions

1. The mean and median of a data set must always be either a positive value or zero.

- A. True
- B. False

Solution: B

Section 13.1 LO 1 Easy

2. The standard deviation of a data set must always be either a positive value or zero.

- A. True
- B. False

Solution: A

Section 13.2 LO 1 Moderate

3. The variance of a data set will always be greater than or equal to the standard deviation of the same data set.

- A. True
- B. False

Solution: B

Section 13.2 LO 3 Moderate

Multiple Choice Questions

1. The annual rates of inflation over a six-year time period are 2.1%, 3.2%, 0.4%, 5.3%, 1.9%, and 4.7%. Determine the arithmetic mean of this data set. Round your answer to two decimal places.

- A. 2.51%
- B. 2.93%
- C. 3.52%
- D. 4.18%

Solution: B. To calculate the arithmetic mean for this data set, add the data values and then divide by the number of data values, which is 6:

$$\bar{x} = \frac{\sum x}{n} = \frac{17.6\%}{6} = 2.93\%$$

Section 13.1 LO 1 Easy

2. The annual returns of six different technology mutual funds are 6.9%, -1.4%, 3.8%, 5.8%, 3.7%, and 8.1%. Determine the median of this data set.
- A. -1.4%
 - B. 2.93%
 - C. 4.8%
 - D. 5.8%

Solution: C. To calculate the median for this data set, order the data from smallest to largest: -1.4%, 3.7%, 3.8%, 5.8%, 6.9%, 8.1%. Because there is an even number of data values, add the two middle values and divide by 2:

$$\text{Median} = \frac{3.8\% + 5.8\%}{2} = \frac{9.6\%}{2} = 4.8\%$$

Section 13.1 LO 1 Moderate

3. A human resource analyst wants to determine the average or typical salary in the company. There are two outlier salaries, those of the CEO and the vice president of the company. What is the best measure to use to represent the typical salary in the company?
- A. Mean
 - B. Median
 - C. Standard deviation
 - D. Interquartile range

Solution: B. The median is the best measure of central tendency when outliers are present in the data set.

Section 13.1 LO 2 Easy

4. A sample is taken of the closing price of the Russell 2000 index over a five-day period. The sample closing prices are 2,240; 2,231; 2,287; 2,210; and 2,198. Calculate the standard deviation of this data set. Round your answer to one decimal place.
- A. 34.4
 - B. 38.4
 - C. 49.2
 - D. 55.1

Solution: A. Using the TI BA II Plus calculator or similar technology, enter the data and follow the steps needed to find the sample standard deviation. For a manual calculation method, use the following formula:

$$s = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

The sample standard deviation is 34.4.

Section 13.2 LO 1 Difficult

5. A sample is taken of the closing price of a technology stock over a six-day period. The sample closing prices are \$177.51, \$178.32, \$173.48, \$176.85, \$175.60, and \$174.95. Calculate the variance of this data set. Round your answer to two decimal places.
- A. 1.78
 - B. 2.88
 - C. 3.18
 - D. 3.88

Solution: C. Using the TI BA II Plus calculator or similar technology, enter the data and execute the steps needed to find the sample variance. For a manual calculation method, use the following formula:

$$s^2 = \frac{\sum(x - \bar{x})^2}{n - 1}$$

The sample variance is 3.18.

Section 13.2 LO 2 Difficult

6. The variance of a data set is 72.25. Determine the standard deviation of the same data set.
- A. 7.225
 - B. 8.5
 - C. 144.5
 - D. 5,220.0625

Solution: B. The standard deviation is calculated as the square root of the variance. In this example, the square root of 72.25 is 8.5, so the standard deviation is 8.5.

Section 13.2 LO 3 Moderate

7. The z-score of the health care–based mutual fund XYZ is known to be -1.5. The mean return of health care mutual funds is 6.8%, and the standard deviation is 0.9%. What is the return for the mutual fund XYZ?
- A. -5.45%
 - B. 1.2%
 - C. 5.45%
 - D. 8.15%

Solution: C. The formula to calculate a z-score is $z = \frac{x - \mu}{\sigma}$, where x is the measurement, μ is the mean, and σ is the standard deviation. In this example, the z-score is known, so the formula can be rewritten as $x = \mu + z\sigma = 6.8\% + (-1.5)(0.9\%) = 5.45\%$.

Section 13.3 LO 1 Difficult

8. The following ordered data set represent the ages of investors at a certain investment firm: 31, 33, 33, 37, 40, 41, 43, 48, 49, 52, 55, 59, 60, 60, 63. Determine the first quartile, median, and third quartile for this data set.
- A. $Q1 = 37, \text{Median} = 48, Q3 = 59$
 - B. $Q1 = 33, \text{Median} = 48, Q3 = 55$
 - C. $Q1 = 40, \text{Median} = 48.5, Q3 = 60$
 - D. $Q1 = 35, \text{Median} = 48, Q3 = 59.5$

Solution: A. The median of the data set is the middle value of the ordered data set, which is 48. The first quartile is the median of the lower half of the data set, which is 37. The third quartile is the median of the upper half of the data set, which is 59.

Section 13.3 LO 2 Moderate

9. The following ordered data set represents the annual returns of six different energy-based mutual funds: -1.2%, 2.3%, 3.5%, 3.8%, 4.2%, 4.4%, 4.6%, 5.1%, 5.7%, 7.6%, 8.0%. Determine the interquartile range (IQR) of this data set.
- A. 2.2%
 - B. 3.5%
 - C. 4.4%
 - D. 5.7%

Solution: A. To calculate the IQR, subtract the first quartile from the third quartile. The median of the data set is the middle value in the ordered data set, which is 4.4%. The first quartile is the median of the lower half of the data set, which is 3.5%. The third quartile is the median of the upper half of the data set, which is 5.7%. The IQR is then calculated as $5.7\% - 3.5\% = 2.2\%$.

Section 13.3 LO 3 Moderate

10. A finance professional plans to create a frequency distribution table using seven classes to summarize portfolio values for clients. The minimum data value is \$32,800, and the maximum data value is \$910,400. Determine the class width for the frequency table. Round your answer up to the next whole number.
- A. 125,371
 - B. 125,372
 - C. 438,000
 - D. 471,600

Solution: B. To calculate the class width, use the following formula:

$$\text{Class Width} = \frac{\text{Max} - \text{Min}}{\text{Number of Classes}} = \frac{910,400 - 32,800}{7} = 125,372$$

Section 13.4 LO 1 Moderate

11. A finance professional has created the frequency distribution table shown below to summarize the portfolio values for 20 clients at a brokerage firm. Determine the relative frequency for the first row in the frequency table.

Portfolio Value Interval (\$000s)	Frequency
0–299	3
300–599	2
600–899	3
900–1,199	6
1,200–1,499	3
1,500–1,799	2
1,800–2,099	1

- A. 0.1
- B. 0.15
- C. 0.2
- D. 0.3

Solution: B. The relative frequency for a certain row is calculated by dividing the frequency for that row by the total number of data values. In this example, the frequency for the first row is 3 and the total number of data values is 20, so the relative frequency is calculated as follows:

$$\text{Relative Frequency} = \frac{3}{20} = 0.15$$

Section 13.4 LO 1 Moderate

12. A human resource analyst creates a frequency distribution table and calculates the frequency for each row in the table. What will be the sum of the frequency column?
- A. The sum will vary depending on the specific data values in the table.
 - B. The sum will equal the number of data values.
 - C. The sum will equal 0.
 - D. The sum will equal 1.

Solution: B. The sum of the frequency column in a frequency distribution table will always equal the number of data values.

Section 13.4 LO 1 Moderate

13. A finance professional has created the frequency distribution table shown below to summarize the portfolio values for 20 clients at a brokerage firm. What is the class width of this frequency table?

Portfolio Value Interval (\$000s)	Frequency
0–250	2
251–501	3
502–752	6
753–1,003	3
1,004–1,254	2
1,255–1,505	3
1,506–1,756	1

- A. 250
- B. 251
- C. 500
- D. 502

Solution: B. The class width is the interval between the lower class limits of two consecutive rows. The lower class limit of the first row is 0, and the lower class limit of the second row is 251. Thus, the class width is $251 - 0 = 251$.

Section 13.4 LO 1 Moderate

14. A bank has measured service time for customers waiting in line, and the service time follows an exponential distribution with a mean waiting time of 1.25 minutes. The bank has a customer service goal that customers should wait in line for less than 2 minutes. Determine the probability that a customer will wait in line for less than 2 minutes. Round your answer to three decimal places.
- A. 0.080
 - B. 0.500
 - C. 0.798
 - D. Not enough information to determine

Solution: C. To calculate the probability, find the area under the exponential distribution to the left of 2 minutes. The mean waiting time is given as 1.25 minutes. The probability can then be calculated as follows:

$$\text{Probability} = 1 - e^{-\frac{1}{1.25}(2)} = 0.798$$

Section 13.4 LO 3 Difficult

15. An investor sets up a retirement portfolio with 60% of the value invested in stocks and 40% of the value invested in bonds. The stocks provide an annual return of 10.8%, and the bonds provide an annual return of 4.3%. Determine the expected value of the return on this portfolio.
- A. 5.00%
 - B. 7.65%
 - C. 8.2%
 - D. 8.75%

Solution: C. Expected Return = $0.60(10.8\%) + 0.40(4.3\%) = 8.2\%$

Section 13.5 LO 2 Moderate

16. The salaries at a financial consulting firm are known to follow a normal distribution. A certain consultant earns a salary that falls exactly at the first quartile of the data set. What is the percentile of this consultant's salary?

- A. 25th percentile
- B. 34th percentile
- C. 50th percentile
- D. 68th percentile

Solution: A. The first quartile corresponds to the 25th percentile in a data set.

Section 13.5 LO 3 Moderate

17. The average annual return on mid-cap stock funds over the five-year period from 2010 to 2015 follows a normal distribution, with a mean rate of return of 8.3% and a standard deviation of 2.2%. Determine the interval for returns that includes approximately 99.7% of the distribution.

- A. 7.2% to 9.4%
- B. 6.1% to 10.5%
- C. 3.9% to 12.7%
- D. 1.7% to 14.9%

Solution: D. According to the empirical rule, 99.7% of a normal distribution falls within three standard deviations of the mean.

$$\text{Lower Bound} = 8.3\% - 3(2.2\%) = 1.7\%$$

$$\text{Upper Bound} = 8.3\% + 3(2.2\%) = 14.9\%$$

Section 13.5 LO 3 Moderate

18. The salaries at a financial consulting firm are known to follow a normal distribution, with mean salary of \$87,000 and standard deviation of \$9,200. Determine the approximate percentage of employees whose salaries fall within the interval from \$87,000 to \$96,200.

- A. 15%
- B. 34%
- C. 68%
- D. 95%

Solution: B. According to the empirical rule, the percentage of the distribution that falls within the interval from the mean to one standard deviation above the mean is 34%.

Section 13.5 LO 3 Moderate

Problems

1. A certain energy-related mutual fund reports a rate of return of 8.2% for a certain year, while the mean rate of return for comparable funds is 6.1% and the standard deviation is 0.68%. Calculate and interpret the z-score for this particular mutual fund.

Solution: To calculate a z-score, use the formula $z = \frac{x - \mu}{\sigma}$, where x is the measurement, μ is the mean, and σ is the standard deviation. In this example, the measurement $x = 8.2$, $\mu = 6.1$, and $\sigma = 0.68$. Using the z-score formula, the calculation is performed as follows:

$$z = \frac{x - \mu}{\sigma} = \frac{8.2 - 6.1}{0.68} = 2.5$$

The resulting z-score of 2.5 represents the number of standard deviations by which a particular measurement is above or below the mean. In this example, the rate of return for this particular mutual fund is 2.5 standard deviations above the mean for comparable funds, indicating that this mutual fund generated a significantly better rate of return compared to the mean rate of return for energy-based mutual funds for the same time period.

Section 13.3 LO 1 Moderate

2. Over a four-year period, the annual rates of return on a certain mutual fund investment are 6.23%, 5.81%, 4.11%, and 3.94%. Determine the single percentage growth rate that is equivalent to these four annual consecutive rates of return. Round your answer to two decimal places.

Solution: The geometric mean is used to find the equivalent percentage growth rate:

$$\tilde{x} = \sqrt[n]{x_1 \cdot x_2 \cdots x_n} = \sqrt[4]{1.0623 \cdot 1.0581 \cdot 1.0411 \cdot 1.0394} = 1.0502$$

The equivalent percentage growth rate is 5.02%.

Section 13.1 LO 3 Difficult

3. The historical returns on the common stock of Acme Manufacturing Corporation over a six-year period are 12%, 8%, 7.5%, -1.5%, 3%, and 4.1%. Determine if there are any outliers in this data set.

Solution: Order the data from smallest to largest: -1.5%, 3%, 4.1, 7.5%, 8%, 12%

Calculate the first and third quartiles: $Q1 = 3\%$, $Q3 = 8\%$

Calculate the IQR: $IQR = Q3 - Q1 = 8\% - 3\% = 5\%$

Calculate the lower and upper bounds for outliers:

$$\text{Lower Bound} = Q1 - (1.5)(IQR) = 3\% - 1.5(5\%) = -4.5\%$$

$$\text{Upper Bound} = Q3 + (1.5)(IQR) = 8\% + 1.5(5\%) = 10.5\%$$

There are no data values below the lower bound, so there are no outliers below the lower bound. There is one data value above the upper bound, which is 12%. Thus, the data value 12% is considered an outlier.

Section 13.3 LO 3 Moderate

4. An investor decides to purchase shares of Omega Manufacturing Corporation on three purchase dates, as shown in the following table. Calculate the weighted mean of the purchase price, where the weights are based on the number of shares in the portfolio. Round your answer to the nearest cent.

Purchase Date	Purchase Price	Number of Shares Purchased
January 31	\$65	500
March 31	\$68	750
July 31	\$51	425

Solution: The weighted mean is calculated by multiplying each purchase price by the number of shares purchased at that price, adding the products together, and dividing the sum by the total number of shares purchased.

$$\text{Weighted Mean} = \frac{\$65(500) + \$68(750) + \$51(425)}{500 + 750 + 425} = \$62.79$$

Section 13.1 LO 3 Moderate

5. The average annual return of the Russell 2000 index is known to follow a normal distribution, with a mean annual return of 6.2% and a standard deviation of 2.3%. What range of values would include 95% of average annual returns for the Russell 2000 index?

Solution: According to the empirical rule, 95% of the data in a normal distribution fall within two standard deviations of the mean. The lower value of this range is two standard deviations below the mean, and the upper value is two standard deviations above the mean.

$$\begin{aligned}\text{Lower Value} &= 6.2\% - 2(2.3\%) = 1.6\% \\ \text{Upper Value} &= 6.2\% + 2(2.3\%) = 10.8\%\end{aligned}$$

Section 13.4 LO 2 Moderate